

HOMEWORK 2

CMU 10-753: DL AND GENAI THEORY

OUT: Friday, Sept 25th, 2026

DUE: Thursday, Oct 8th, 2026, 11:59pm

START HERE: Instructions

- **Collaboration policy:** Collaboration on solving the homework is allowed, after you have thought about the problems on your own. To remind you, many questions in this HW have solutions that are very easy to find online (and many are from previous versions of this course). It is also OK to get clarification (but not solutions) from books or online resources, again after you have thought about the problems on your own. There are two requirements: first, cite your collaborators fully and completely (e.g., “Jane explained to me what is asked in Question 2.1”). Second, write your solution *independently*: close the book and all of your notes, and send collaborators out of the room, so that the solution comes from you only.
- **Submitting your work:**
 - **Gradescope:** For the written problems such as short answer, multiple choice, derivations, proofs, or plots, we will be using the Gradescope. The best way to format your homework is by using the Latex template released in the handout and writing your solutions in Latex. However, submissions can be handwritten onto the template, but should be labeled and clearly legible. If your writing is not legible, you will not be awarded marks.
Regrade requests can be made after the homework grades are released, however this gives the TA the opportunity to regrade your entire paper, meaning if additional mistakes are found then points will be deducted.
 - **Programming:** You should submit all code used to solve the programming aspect of the homework to the corresponding ‘Programming’ submission slot on Gradescope. If you do not do this, you will not get any credit for any of the programming section irrespective of the plots and values submitted to the ‘Written’ submission slot.

1 Ito Integral Questions - Saumya (20 Points)

Problem 1. Find the distribution of

$$X = \int_0^2 e^{-t} dW_t.$$

Problem 2. Find the distribution of

$$X = \int_0^{1/2} dW_t + 2 \int_{1/2}^1 dW_t.$$

Problem 3. Let

$$X = \int_0^1 t dW_t, \quad Y = \int_0^1 (1-t) dW_t.$$

Find the joint distribution of (X, Y) . In particular, calculate

$$\text{Var}(X), \quad \text{Var}(Y), \quad \text{Cov}(X, Y).$$

Are X and Y independent?

HINT: Use the polarization form of the Ito isometry:

$$\text{Cov}(X_T, Y_T) = \text{Cov} \left(\int_0^T f(t) dW_t, \int_0^T g(t) dW_t \right) = \int_0^T f(t)g(t) dt.$$

Problem 4. Let

$$X = \int_0^1 1 dW_t, \quad Y = \int_0^1 \left(t - \frac{1}{2} \right) dW_t.$$

Show that X and Y are independent.

Problem 5. Use Ito's formula to express

$$\int_0^1 W_t^2 dW_t$$

in terms of W_1 and an ordinary time integral.

Ito's formula:

$$df(W_t) = f'(W_t) dW_t + \frac{1}{2} f''(W_t) dt$$

Problem 6. Without explicitly finding its distribution, calculate

$$\mathbb{E} \left[\int_0^1 W_t dW_t \right]$$

and

$$\mathbb{E} \left[\left(\int_0^1 W_t dW_t \right)^2 \right].$$

Problem 7. Show that

$$\int_0^1 t \, dW_t = W_1 - \int_0^1 W_t \, dt.$$

HINT: Use the stochastic product rule

$$d(X_t Y_t) = X_t \, dY_t + Y_t \, dX_t + d[X, Y]_t.$$

Use this identity to calculate

$$\text{Var} \left(\int_0^1 W_t \, dt \right).$$

Problem 8. Let

$$X = \int_0^{1/2} t \, dW_t, \quad Y = \int_{1/2}^1 t \, dW_t.$$

Find the distributions of X , Y , and $X + Y$. Are X and Y independent?

Problem 9. For $s \in [0, 1]$, define

$$X_s = \int_0^s t \, dW_t.$$

Find

$$\mathbb{E}[X_s], \quad \text{Var}(X_s), \quad \text{Cov}(X_s, X_t).$$

Is $(X_s)_{s \in [0, 1]}$ a Gaussian process?

Problem 10. Determine whether each statement is true or false and justify your answer.

- (a) $\int_0^1 f(t) \, dW_t$ is Gaussian for every deterministic $f \in L^2([0, 1])$.
- (b) $\int_0^1 X_t \, dW_t$ is Gaussian for every square-integrable adapted process X_t .
- (c) $\mathbb{E} \left[\int_0^1 X_t \, dW_t \right] = 0$ for every square-integrable adapted process X_t .
- (d) $\text{Var} \left(\int_0^1 f(t) \, dW_t \right) = \int_0^1 f(t)^2 \, dt$ for every deterministic $f \in L^2([0, 1])$.

2 Add a Drift, Keep the Roughness -Ibrahim (20 Points)

Recall from lecture that a real trajectory is often modeled as a smooth deterministic path plus a random perturbation, $X_t = f(t) + W_t$. Lecture already established that these two pieces behave at opposite extremes, for continuously differentiable f ,

$$TV(f, [0, T]) = \int_0^T |f'(t)| \, dt < \infty$$

and $[f]_T = 0$, while a standard Wiener process satisfies

$$TV(W, [0, T]) = \infty$$

almost surely and $Q_n(W) := \sum_{i=0}^{n-1} (W_{t_{i+1}} - W_{t_i})^2 \xrightarrow{L^2} T$ along the uniform partition $\Pi_n = \{t_0, \dots, t_n\}$, $t_i = i\Delta t$, $\Delta t = T/n$.

You may use, without proof, the Mean Value Theorem bound $|f(t_{i+1}) - f(t_i)| \leq M\Delta t$, where $M := \max_{[0, T]} |f'| < \infty$.

For this question, let $f : [0, T] \rightarrow \mathbb{R}$ be continuously differentiable and deterministic, and define $X_t := f(t) + W_t$.

Question: Show that $Q_{\Pi_n}(X) := \sum_{i=0}^{n-1} (X_{t_{i+1}} - X_{t_i})^2 \xrightarrow{L^2} T$.

Question: Show that $TV(X, [0, T]) = \infty$ almost surely.

Note: You are **not** being asked to reprove $TV(W, [0, T]) = \infty$, that fact is given directly. The quadratic-variation-based contradiction argument used in lecture to establish the fact about W is not needed or relevant here.

3 Programming Question: Wiener Process - Krish (20 points)

Your task is to implement an approximation of a Wiener process on $[0, 1]$ and study some of its properties. We provide a Jupyter notebook file with a few missing cells https://github.com/bapoczko/10753/blob/main/Wiener_Process_Demo_Question.ipynb. Please follow the instructions in this notebook and complete the missing parts.

You will need to submit your completed Jupyter notebook. In this notebook, you will need to implement the:

- `simulate_wiener` function (5 points)
- `print_table` function (5 points)
- `plot_paths` function (5 points)
- `calc_increments` function (5 points)

As a proof-of-execution check, add the following figures directly below this question in this document (not just in your notebook). Each figure must have a caption that clearly indicates which figure is being shown. You will separately submit your completed Jupyter notebook (`.ipynb`) as well.

1. **output of the `print_table` function:** print the empirical and theoretical mean and variance values as a table, for time points $[0.1, 0.25, 0.5, 0.75, 1.0]$. An example output is shown in the notebook file

2. **output of the plot_paths function:** Plot 25 generated sample paths from the Wiener process. Also plot the 95% envelope, that is the $1.96 * \text{np.sqrt}(t)$ and $-1.96 * \text{np.sqrt}(t)$ curves that contain 95% of the Gaussian trajectories at any t . An example is shown in the notebook file.
3. **output of the calc_increments function:** calculate the $W(a) - W(b)$ increments for $a = 0.7$ and $b = 0.3$. Print their empirical and theoretical mean and variance. Plot a histogram of the $W(a) - W(b)$ increments Plot the theoretical Gaussian pdf of $W(a) - W(b)$. An example is shown in the notebook file.

4 Programming Question: Stochastic Integrals - Krish (25 points)

The goal of this HW exercise is to demonstrate the different types of stochastic integrals as a limit of Riemann-Stieltjes-type sums.

We are going to use the uniform partition of $[0,1]$: $t_i = i/n, i = 0, \dots, n$, and we will also use the discrete Wiener approximation $W^{(n)}$ from the 'simulate_wiener' function implemented in the previous exercise.

For each partition $[t_i, t_{i+1}]$, we will select a $c_i \in [t_i, t_{i+1}]$. We will investigate three cases when this c_i is the left, mid, or right endpoint of each subinterval. We call this selection 'tag', so $\text{tag} \in \{\text{'left'}, \text{'mid'}, \text{'right'}\}$.

You will need to calculate the Riemann-Stieltjes-type sums:

$$S_n(f, \text{tag}) = \sum_{i=0}^{n-1} f(c_i) * (W_{t_{i+1}} - W_{t_i}), \quad c_i \in [t_i, t_{i+1}]$$

on the partition $t_i = i/n, i = 0, \dots, n$ of $[0, 1]$, using the wiener_approximation function you implemented.

We will investigate two different functions.

(A): $f(t) = t$, so the integral is $I_1 = \int_0^1 f(t) dW(t) = \int_0^1 t dW(t)$. Here f is deterministic.

(B): $f(t) = W(t)$, so the integral is $I_2 = \int_0^1 f(t) dW(t) = \int_0^1 W(t) dW(t)$. Here f is stochastic.

Task 1 (2 Points)

Argue that $I_1 = \int_0^1 t dW(t) \sim \mathcal{N}(0, 1/3)$ that is I_1 has a Gaussian distribution with zero mean and $1/3$ variance. You can use any of the theorems stated in class.

Task 2 (5 Points)

With $n = 5000$ grid points, simulate $m=10,000$ Wiener process trajectories (W_t). For each of these trajectories evaluate $S_n^m(f, \text{tag})$ with $\text{tag}=\text{'left'}$. Plot the histogram of these $m=10,000$

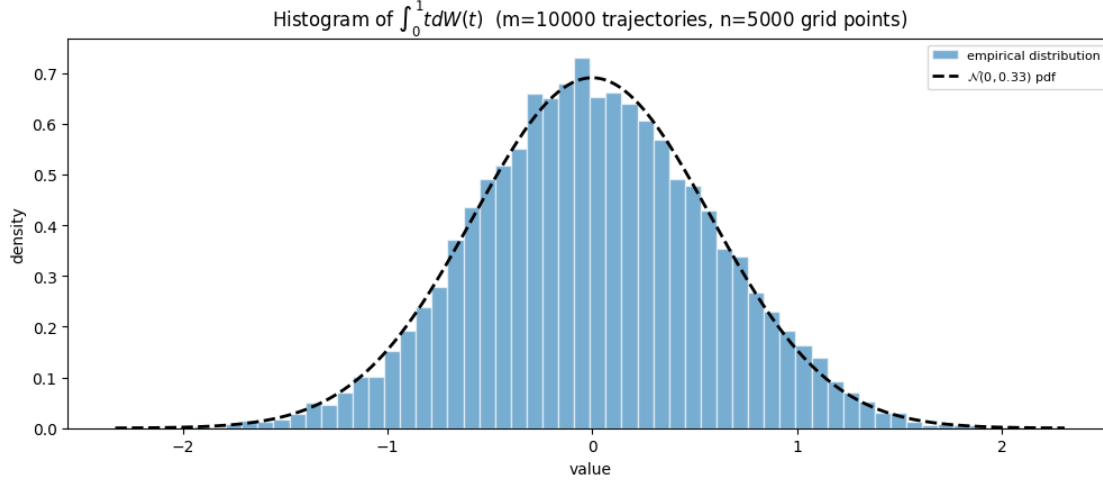


Figure 1: Histogram of $\int_0^1 t dW(t)$. We can see that the histogram is quite close to $\mathcal{N}(0, 1/3)$

points and plot also the theoretical pdf.

On Figure 1 we show a possible solution.

Task 3 (5 Points)

Let us demonstrate that since $f(t) = t$ is deterministic, it doesn't matter what tag we choose, the integral values will be the same for large n . For each of the m Wiener process trajectories, using n grid points, calculate $S_n^m(f, \text{'left'})$, $S_n^m(f, \text{'mid'})$, $S_n^m(f, \text{'right'})$ quantities.

Plot the below values for for $n = [10, 20, 50, 100, 200, 500, 1000, 2000, 5000]$:

$$\max_m |S_n^m(f, \text{'left'}) - S_n^m(f, \text{'mid'})|$$

$$\max_m |S_n^m(f, \text{'left'}) - S_n^m(f, \text{'right'})|$$

On Figure 2, we show a possible solution. As you can see these differences converge to 0 as $n \rightarrow \infty$.

Task 4 (5 Points)

We will use $f(t) = W(t)$ stochastic function in this task.

Similarly to Task 2, show the histogram of the estimation of $I_2 = \int_0^1 W(t) dt$. Use $n = 5000$ grid points, $m=10,000$ Wiener process trajectories (W_t). For each of these trajectories, evaluate $S_n^m(f, \text{tag})$ with $\text{tag}=\text{'left'}$. Plot the histogram of these $m=10,000$ points.

Figure 3 shows a possible solution.

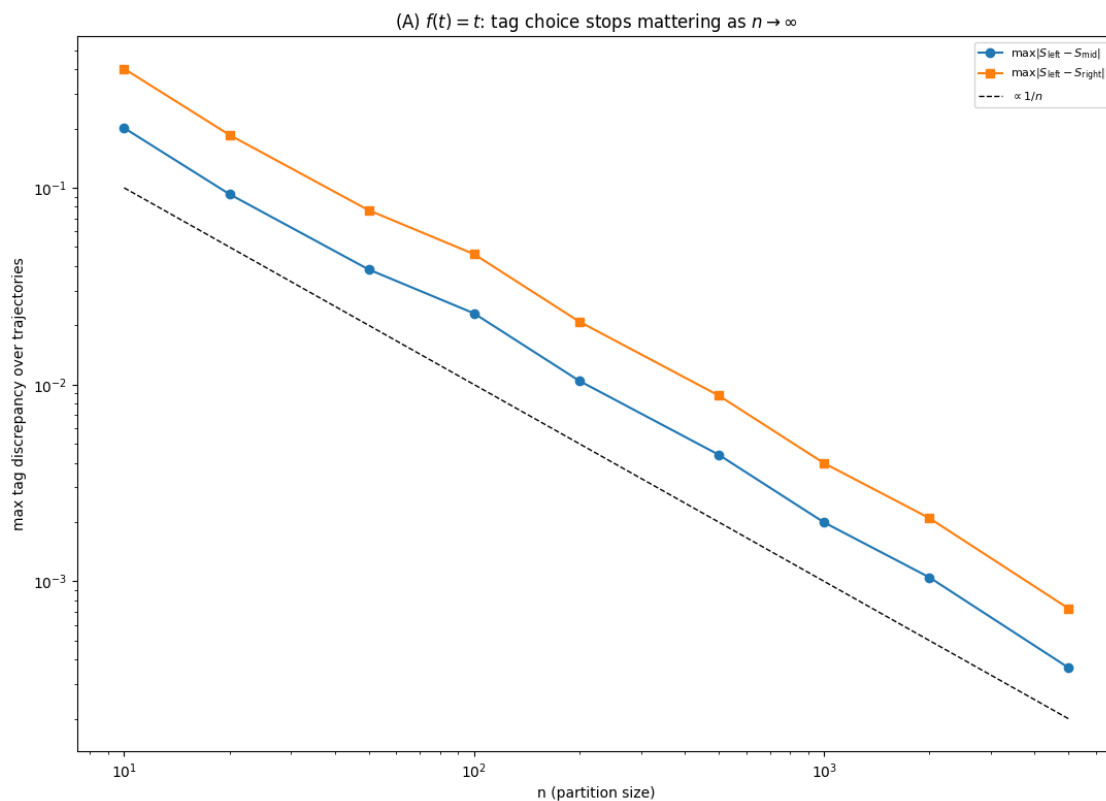


Figure 2: Tag choice doesn't matter for deterministic functions when n is large

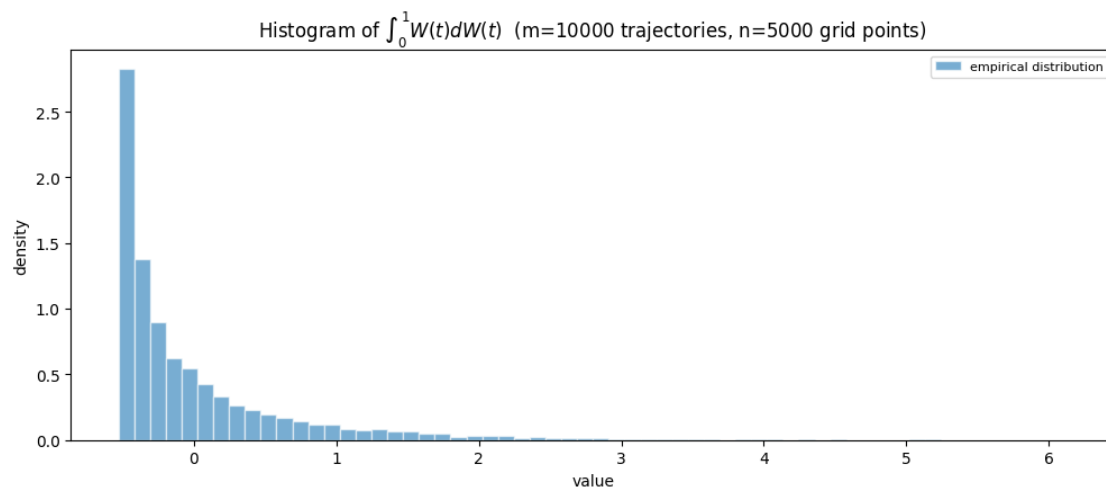


Figure 3: Histogram of $\int_0^1 W(t) dW(t)$. We can see that the histogram is not Gaussian distribution

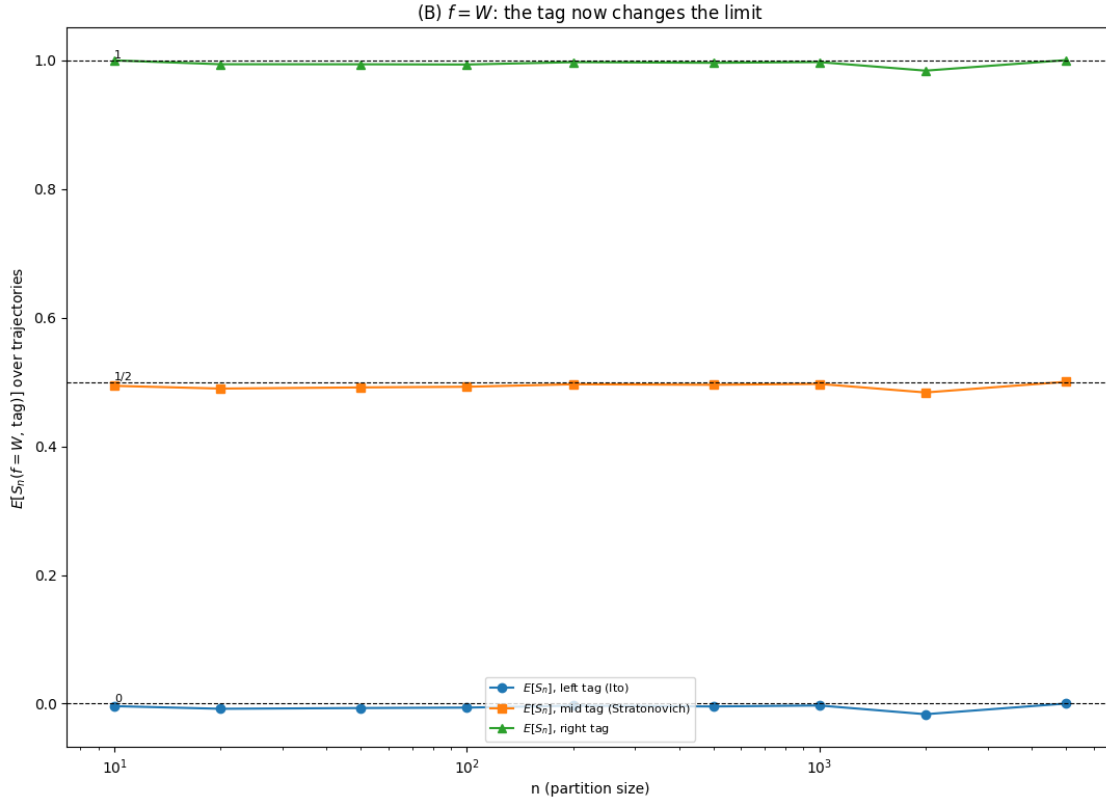


Figure 4: Tag choice does matter for stochastic functions when n is large

Task 5 (5 Points)

Let us repeat the task of Task 3, but using the $f(t)=W(t)$ function, that is by approximating $I_2 = \int_0^1 W(t) dW(t)$ integral. Demonstrate that the tag choice matters in this case.

A possible solution is on Figure 4.

Important!

Important: Remember that you **MUST** submit all code used in this part to the Programming submission slot on Gradescope otherwise you will not get credit for this section.

Before generating any figures below or submitting, restart the kernel and run all cells top to bottom (Kernel/Runtime → Restart & Run All) so the notebook you submit reproduces cleanly from a blank state. Submit this restarted, fully-run notebook (.ipynb) along with your write-up.

5 Collaboration Questions

1. (a) Did you receive any help whatsoever from anyone in solving this assignment?
(b) If you answered ‘yes’, give full details (e.g. “Jane Doe explained to me what is asked in Question 3.4”)

2. (a) Did you give any help whatsoever to anyone in solving this assignment?
(b) If you answered ‘yes’, give full details (e.g. “I pointed Joe Smith to section 2.3 since he didn’t know how to proceed with Question 2”)

3. (a) Did you find or come across code that implements any part of this assignment?
(b) If you answered ‘yes’, give full details (book & page, URL & location within the page, etc.).